ARTIFICIAL INTELLIGENCE

Student Study Material

Structured notes • Core concepts • Examples • Revision tables • Practice questions

Designed for: Beginner students and introductory AI / ML coursework

A clear foundation for understanding what AI is, how it works, where it is used, and what responsible AI requires.

How to use these notes

- Read each chapter in order, starting with the basic vocabulary.
- Use the comparison tables to revise similar concepts.
- Try the review questions at the end without looking at the notes.
- Check your course syllabus: this is a general introductory guide, not a replacement for your lecturer's material.

Contents

1. Introduction to Artificial Intelligence
2. History and Types of AI
3. Intelligent Agents and Problem Solving
4. Search Strategies
5. Knowledge Representation and Reasoning
6. Machine Learning: The Connection to AI
7. Neural Networks and Deep Learning
8. Natural Language Processing
9. Computer Vision, Robotics and Other Applications
10. Generative AI and Large Language Models
11. AI Development Workflow and Evaluation
12. Ethics, Safety, Privacy and Responsible AI
13. Key Terms and Quick Revision
14. Practice Questions with Answer Guide
15. Mini Glossary and Study Plan

1. Introduction to Artificial Intelligence

1.1 What is Artificial Intelligence?

Artificial Intelligence (AI) is a field of computing concerned with building systems that can perform tasks commonly associated with human intelligence. These tasks may include learning from data, recognizing patterns, understanding language, planning, solving problems and making recommendations.

AI does not mean that a computer thinks or feels exactly like a human. An AI system uses algorithms, data and computing resources to produce outputs such as predictions, classifications, plans or generated content.

1.2 Main goals of AI

- **Learning:** improve performance using examples or experience.
- **Reasoning:** use facts and rules to reach conclusions.
- **Problem-solving:** search for a solution among possible choices.
- **Perception:** interpret inputs such as images, audio or sensor readings.
- **Language:** process or generate human language.
- **Planning and action:** choose steps to reach a goal.

1.3 AI, automation and traditional programming

Approach	How it works	Example
Traditional programming	A programmer writes explicit rules; input is processed by those rules.	A calculator applying a formula.
Automation	A workflow carries out predefined steps, often without learning.	A light turns on at a set time.
AI system	Uses rules, learned patterns, search or a combination to produce an intelligent-seeming result.	An email filter identifying likely spam.

1.4 Common characteristics

- Works toward a defined task or objective.
- Processes input and produces an output.
- May use rules, search, statistical models or learned representations.
- Can be uncertain or wrong, especially when inputs differ from training examples.
- Usually needs human design, testing, monitoring and appropriate oversight.

Example: A student-support chatbot receives a question, processes the words, uses a language model or knowledge source, and generates a response. The response should be checked for accuracy when the stakes are important.

2. History and Types of AI

2.1 A short history

Period / milestone	Why it matters
1940s–1950s	Early computing and foundational ideas about machine intelligence developed.
1956	The Dartmouth summer research project helped establish AI as a named research field.
1960s–1970s	Researchers explored symbolic reasoning, search, games and early language systems.
1970s–1990s	Periods of reduced funding and expectations became known as AI winters; expert systems also gained attention.
1990s–2000s	Statistical machine learning became increasingly important; systems achieved notable task-specific results.
2010s	Large datasets, GPUs and deep learning improved image, speech and language tasks.
2020s	Generative AI and large language models became widely accessible, while concerns about reliability and governance grew.

2.2 Types by capability

Type	Meaning	Current status
Narrow AI (ANI)	Designed for specific tasks or a limited range of tasks.	Common today: translation, recommendations, image recognition.
Artificial General Intelligence (AGI)	A hypothetical system able to learn and perform a broad range of intellectual tasks at a human-like general level.	No agreed demonstration of AGI; definitions vary.
Artificial Superintelligence (ASI)	A hypothetical AI that exceeds human capabilities across most or all relevant intellectual domains.	Speculative concept.

2.3 Types by approach

- **Rule-based / symbolic AI:** represents knowledge with symbols and explicit rules.
- **Machine learning:** learns patterns from data instead of relying only on hand-written rules.
- **Hybrid AI:** combines learned models with rules, databases or other reasoning methods.

Important distinction: 'narrow, general and superintelligent' describes breadth of capability, while 'symbolic, machine learning and hybrid' describes approaches. These are different classification schemes.

3. Intelligent Agents and Problem Solving

3.1 What is an intelligent agent?

An **agent** is a system that observes its environment through inputs or sensors and acts on it through outputs or actuators. An intelligent agent selects actions in a way intended to achieve its objectives.

Agent cycle: perceive → interpret / decide → act → observe the result → repeat.

3.2 PEAS description

PEAS is a framework for specifying a task environment:

Letter	Meaning	Example: delivery robot
P	Performance measure: what counts as success	Deliver correctly, safely and on time.
E	Environment: where the agent operates	Roads, pavements, people, weather.
A	Actuators: how it acts	Wheels, brakes, robotic arm.
S	Sensors: how it observes	Cameras, GPS, proximity sensors.

3.3 Task environment properties

- **Fully observable vs partially observable:** whether the agent can access all relevant state information.
- **Deterministic vs stochastic:** whether an action's outcome is predictable or involves uncertainty.
- **Episodic vs sequential:** whether each decision is independent or affects later decisions.
- **Static vs dynamic:** whether the environment can change while the agent decides.
- **Discrete vs continuous:** whether states/actions take distinct values or vary over a range.
- **Single-agent vs multi-agent:** whether one or several agents affect the outcome.

3.4 Problem formulation

A search problem is often described using an initial state, available actions, a transition model, a goal test and a path-cost function. For example, in route planning, the initial state is the starting city, actions are road choices, the goal is the destination and cost may represent distance or travel time.

1. Define the problem and goal clearly.
2. Describe the possible states and actions.
3. Specify how actions change the state.
4. Choose a goal test and cost measure.
5. Select an algorithm and evaluate its solution.

4. Search Strategies

4.1 Why search is needed

Many AI problems have too many possible choices to inspect manually. Search algorithms systematically explore candidate states or paths to find a goal or a good solution.

4.2 Uninformed search

Algorithm	Core idea	Strength / limitation
Breadth-First Search (BFS)	Explores shallowest nodes first using a queue.	Finds a shortest path when each step has equal cost; may use much memory.
Depth-First Search (DFS)	Explores one branch deeply before backtracking, often using a stack.	Memory-efficient in some settings; can follow a long or infinite branch.
Uniform-Cost Search (UCS)	Expands the frontier path with the lowest accumulated cost.	Can find a least-cost solution with suitable non-negative costs; may explore many states.

4.3 Informed search

Informed search uses a **heuristic**: an estimate of how far a state is from the goal.

- **Greedy Best-First Search:** prioritizes the state with the most promising heuristic estimate. It can be fast but is not always optimal.
- **A* Search:** prioritizes f(n) = g(n) + h(n), where g(n) is the cost so far and h(n) estimates the remaining cost. Under suitable conditions, A* is complete and optimal.
- **Heuristic quality:** an admissible heuristic does not overestimate the true remaining cost. Consistency is a stronger useful property.

4.4 Mini example

Imagine finding a route from College A to College B. BFS is suitable when every road segment counts equally. UCS is useful when roads have different costs. A* can use straight-line distance as a heuristic to guide the search toward the destination.

4.5 Search comparison

Question to ask	Why it matters
Does it find a solution if one exists?	Completeness
Does it find the lowest-cost solution?	Optimality
How many states may it explore?	Time complexity
How much memory does it need?	Space complexity

5. Knowledge Representation and Reasoning

5.1 Knowledge representation

Knowledge representation (KR) is the way an AI system stores information about objects, relationships, events or rules so it can use that information to answer questions or make decisions.

- **Logic:** represents facts and formal relationships.
- **Rules:** IF condition THEN conclusion or action.
- **Semantic networks / knowledge graphs:** represent entities and relationships.
- **Ontologies:** define concepts and relationships within a domain.
- **Probabilistic representations:** express uncertainty using probabilities.

5.2 Rule-based reasoning example

Rule: IF a device is plugged in AND the power switch is on, THEN it may receive power.
Facts: the device is plugged in; the switch is on.
Inference: the system concludes that the device may receive power, though a fault could still prevent it.

5.3 Deduction, induction and abduction

Reasoning type	Meaning	Example
Deduction	Apply a general rule to specific facts.	All mammals are warm-blooded; a whale is a mammal; therefore a whale is warm-blooded.
Induction	Infer a general pattern from examples.	Many observed metal objects expand when heated; infer that heating often expands metals.
Abduction	Suggest a plausible explanation for an observation.	The floor is wet; perhaps water was spilled.

5.4 Limits

Knowledge bases can be incomplete or outdated. Rules may conflict, and uncertain real-world evidence can make conclusions unreliable. Systems should indicate uncertainty and be updated when facts change.

6. Machine Learning: The Connection to AI

6.1 What is machine learning?

Machine Learning (ML) is a branch of AI in which a model learns patterns from data to make predictions, classify examples, discover structure or support decisions.

A model is trained by adjusting its internal parameters to perform a task according to an objective or learning procedure.

6.2 Main learning types

Type	Training information	Example
Supervised learning	Examples include inputs and correct target labels / values.	Predict house prices; classify emails as spam or not spam.
Unsupervised learning	Data has no supplied target labels; model looks for structure.	Group customers by similar behaviour.
Reinforcement learning	An agent receives rewards or penalties from actions in an environment.	Train an agent to play a game.
Self-supervised learning	The data itself provides a learning signal, often by hiding or predicting parts of the input.	Train a language model to predict the next token.

6.3 Basic ML workflow

1. Collect data with appropriate permission and quality checks.
2. Clean and prepare data; handle missing values and inconsistent formats.
3. Split data into training, validation and test sets where appropriate.
4. Choose a model and train it on the training data.
5. Tune choices using validation data.
6. Evaluate once on suitable test data and examine errors, not only one score.
7. Deploy carefully, monitor performance and update when needed.

6.4 Important terms

- **Feature:** an input variable used by a model.
- **Label / target:** the value a supervised model is meant to predict.
- **Training:** fitting model parameters to data.
- **Inference:** using a trained model to produce an output.
- **Overfitting:** model learns training details too closely and performs poorly on new data.
- **Underfitting:** model is too simple or insufficiently trained to capture important patterns.
- **Data leakage:** information unavailable at real prediction time improperly enters training or evaluation.

6.5 Classification and regression

Task	Output	Example metrics
Classification	A category or class, such as spam / not spam.	Accuracy, precision, recall, F1-score; ROC-AUC in suitable settings.
Regression	A numerical value, such as tomorrow's temperature.	MAE, MSE / RMSE, R^2, chosen according to the task.

Warning: Accuracy alone can be misleading when classes are imbalanced. For a medical screening model, for example, false negatives and false positives can have very different consequences.

7. Neural Networks and Deep Learning

7.1 What is a neural network?

An artificial neural network is a model made of connected computational units often called neurons. A typical network contains an input layer, one or more hidden layers and an output layer.

7.2 How a neuron works

A simplified neuron computes a weighted combination of inputs, adds a bias and applies an activation function:
z = w■x■ + w■x■ + … + b
output = activation(z)
Weights determine the influence of inputs; bias shifts the calculation; the activation function allows the network to represent nonlinear patterns.

7.3 Training overview

1. Feed training examples through the network (forward pass).
2. Compare predictions with target values using a loss function.
3. Use backpropagation to calculate how parameters contributed to the error.
4. Use an optimizer such as gradient descent to update parameters.
5. Repeat over many examples and monitor validation performance.

7.4 Deep learning

Deep learning uses neural networks with multiple layers to learn complex representations. It has been especially influential in image recognition, speech, language and generative AI.

7.5 Common architecture families

Family	Often associated with
Convolutional Neural Networks (CNNs)	Images and spatial patterns.
Recurrent Neural Networks (RNNs)	Sequential data; many tasks now use other architectures.
Transformers	Attention-based processing of sequences; widely used in language and multimodal models.
Autoencoders	Learning compressed representations and reconstructing inputs.
Generative Adversarial Networks (GANs)	Generating data through competing generator and discriminator networks.

7.6 Strengths and limitations

- Can learn complex patterns from large datasets.
- May need substantial data, compute and energy.
- Can inherit bias in training data.
- May be difficult to interpret.
- Strong performance on one benchmark does not guarantee safe real-world behaviour.

8. Natural Language Processing

8.1 What is NLP?

Natural Language Processing (NLP) is the area of AI that enables computers to analyze, interpret or generate human language in text or speech.

8.2 Common NLP tasks

- **Tokenization:** splitting text into units such as words, subwords or punctuation.
- **Sentiment analysis:** estimating expressed sentiment or opinion.
- **Text classification:** assigning categories to text.
- **Named entity recognition:** identifying names, places, organizations and other entities.
- **Machine translation:** translating between languages.
- **Summarization:** producing a shorter version of a text.
- **Question answering:** generating or retrieving answers to questions.
- **Speech recognition:** converting speech to text; text-to-speech generates spoken audio.

8.3 Language models

A language model estimates patterns in language. Many modern models are trained to predict the next token or fill in missing text. During generation, they use learned patterns and a decoding procedure to produce a sequence of tokens.

8.4 Why NLP is difficult

- Words can have multiple meanings depending on context.
- Sarcasm, idioms, dialects and cultural references can be difficult.
- A sentence may be grammatically correct but factually wrong.
- Training data may underrepresent some languages or communities.
- Private or sensitive text must be handled carefully.

9. Computer Vision, Robotics and Other Applications

9.1 Computer vision

Computer vision enables systems to extract information from images and videos.

- **Image classification:** assign a label to an image.
- **Object detection:** locate and classify objects in an image.
- **Segmentation:** label pixels or regions.
- **OCR:** recognize printed or handwritten text in images.
- **Pose estimation:** estimate the position of body joints or objects.

9.2 Robotics

Robotics combines sensing, control, planning and physical action. AI can help a robot recognize objects, navigate, plan routes or adapt to changing conditions. Not every robot is AI-powered; many perform fixed programmed motions.

9.3 Applications across sectors

Sector	Potential uses	Important caution
Education	Practice feedback, tutoring support, translation, accessibility.	Check accuracy; protect student data; keep human teaching central.
Healthcare	Image support, risk prediction, administrative assistance.	Clinical validation, privacy, bias checks and professional oversight.
Agriculture	Crop monitoring, pest detection, irrigation support.	Local conditions and sensor quality affect results.
Transport	Route planning, traffic prediction, driver-assistance systems.	Safety testing and clear human responsibility.
Finance	Fraud detection, forecasting support, document processing.	Fairness, explainability and regulatory requirements.
Manufacturing	Quality inspection, predictive maintenance, process optimization.	Monitor failures and protect workers.
Environment	Weather analysis, biodiversity monitoring, energy optimization.	Models and sensors have uncertainty; validate findings.

10. Generative AI and Large Language Models

10.1 Generative AI

Generative AI creates new content—such as text, images, audio, video or code—based on patterns learned during training and the prompt or context supplied at use time.

10.2 Large language models (LLMs)

An LLM is a language model trained on very large amounts of text or other tokenized data. Many use transformer architectures. They can summarize, draft, explain, classify, translate and assist with coding, but their outputs are not guaranteed to be true.

10.3 Prompt engineering

Prompt engineering is the practice of writing and refining instructions so a model is more likely to produce a useful output.

- **Role / context:** describe the perspective or setting when useful.
- **Task:** state exactly what should be done.
- **Audience:** specify who the result is for.
- **Constraints:** define length, tone, language, format and boundaries.
- **Output format:** request a table, outline, checklist or structured notes.
- **Verification:** ask for assumptions and sources when available, then check important claims independently.

> **Example prompt:** “Explain supervised and unsupervised learning for a first-year student. Use simple English, one everyday example for each, a comparison table, and five revision questions. State any limitations clearly.”

10.4 Limitations and risks

- **Hallucination:** a model may confidently generate incorrect or invented information.
- **Bias:** outputs may reproduce stereotypes or unfair patterns.
- **Privacy:** avoid entering confidential, identifying or sensitive data into tools unless approved.
- **Copyright and attribution:** respect source rights and course rules.
- **Overreliance:** generated answers should support, not replace, learning and judgment.
- **Security:** malicious prompts or untrusted content can attempt to influence system behaviour.

10.5 Responsible student use

- Use AI to clarify concepts, create practice questions and get feedback.
- Verify facts using textbooks, lecture notes and trustworthy sources.
- Do your own reasoning and follow your institution's academic-integrity policy.
- Do not submit generated work as your own if the rules prohibit it.
- Never assume a generated citation or reference is real without checking it.

11. AI Development Workflow and Evaluation

11.1 End-to-end workflow

1. Define the real problem, users, risks and success criteria.
2. Check whether AI is necessary; a simple rule or conventional program may be better.
3. Collect representative data lawfully and document its limitations.
4. Prepare data and establish a simple baseline.
5. Select and train a suitable approach.
6. Evaluate using metrics that reflect real-world costs and user needs.
7. Test for robustness, bias, security, privacy and failure modes.
8. Deploy with appropriate human oversight and clear user communication.
9. Monitor changes in data and performance; maintain, update or retire the system.

11.2 Train, validation and test data

Training data is used to fit a model. Validation data helps compare models or tune settings. Test data estimates final performance on examples not used for model selection. Repeatedly tuning against the test set weakens its role as an independent check.

11.3 Evaluation concepts

Metric / concept	Meaning
Accuracy	Fraction of predictions that are correct.
Precision	Among predicted positives, the fraction that are truly positive.
Recall (sensitivity)	Among actual positives, the fraction correctly identified.
F1-score	Harmonic mean of precision and recall.
MAE	Average absolute difference between predicted and actual numeric values.
Robustness	Ability to maintain useful performance under reasonable changes or disturbances.
Fairness assessment	Checks whether performance or outcomes differ unfairly across relevant groups.

11.4 Example: spam detection

Suppose an email model marks 100 messages as spam and 80 really are spam. Its precision is 80/100 = 80%. If there were 200 spam messages in total and it found 80 of them, its recall is 80/200 = 40%. This example shows why a single metric may not describe the full performance picture.

12. Ethics, Safety, Privacy and Responsible AI

12.1 Key principles

- **Fairness:** examine whether the system creates or reinforces unjustified differences in outcomes.
- **Transparency:** communicate when AI is used and explain important limits.
- **Accountability:** identify who is responsible for decisions, monitoring and correction.
- **Privacy:** collect only needed data, protect it and respect consent and applicable law.
- **Safety and reliability:** test foreseeable failures and reduce harmful outcomes.
- **Human oversight:** provide meaningful review and a way to challenge consequential decisions.
- **Accessibility:** consider users with different languages, abilities, devices and levels of access.

12.2 Common risks

Risk	Illustration	Possible mitigation
Bias	A hiring model performs worse for one group.	Audit representative data and outcomes; involve affected groups.
Privacy breach	Sensitive records are exposed.	Minimize data, control access, encrypt and define retention.
Misinformation	A chatbot invents a source or fact.	Verify claims, show uncertainty and use reliable retrieval sources.
Automation bias	People accept a model's output without checking it.	Train users, show limitations and require review for high-impact decisions.
Security abuse	Inputs manipulate a system or expose protected information.	Threat-model, test adversarial inputs and restrict access.
Environmental cost	Training and operation consume energy and hardware.	Measure resource use and choose efficient methods where suitable.

12.3 A practical checklist

- What problem is being solved, and who benefits?
- What data was used, and was it collected appropriately?
- Who could be harmed if the system is wrong?
- How will errors be detected and corrected?
- Can a person question or appeal an important decision?
- Is the system monitored after deployment?
- Is AI the right solution, or would a simpler method be safer?

13. Key Terms and Quick Revision

Term	Short definition
Algorithm	A defined procedure for solving a problem or completing a task.
Model	A learned or specified representation used to produce predictions or outputs.
Dataset	A collection of examples or records used for analysis or training.
Parameter	A value adjusted during model training.
Feature	An input variable or representation used by a model.
Inference	Applying a trained model to new input.
Heuristic	A practical estimate or rule used to guide search or decisions.
Agent	A system that observes and acts in an environment.
Neural network	A model of connected computational units with adjustable parameters.
Deep learning	Machine learning based on neural networks with multiple layers.
NLP	AI methods for working with human language.
Computer vision	AI methods for interpreting images and video.
Generative AI	AI that creates content such as text, images, audio or code.
Hallucination	Generated content that appears plausible but is false or unsupported.
Overfitting	Learning training-specific patterns that do not generalize well.
Bias	A systematic tendency that can produce inaccurate or unfair outcomes.
Explainability	How well people can understand factors behind a model's output.

13.1 Quick comparison

Compare	Main difference
AI vs ML	AI is the broader field; ML is one approach within AI.
ML vs deep learning	Deep learning is a subset of ML using multi-layer neural networks.
Supervised vs unsupervised	Supervised uses target labels; unsupervised looks for structure without supplied labels.
Classification vs regression	Classification predicts categories; regression predicts numerical values.
Search vs learning	Search explores possible states; learning fits patterns or policies from data/experience.
Generative vs discriminative	Generative methods model or create data; discriminative methods often predict labels or distinguish classes. Exact meanings depend on context.

14. Practice Questions with Answer Guide

14.1 Short-answer questions

1. Define Artificial Intelligence in your own words.
2. Give four tasks that AI systems can perform.
3. Distinguish traditional programming from machine learning.
4. What is an intelligent agent? Explain the perceive–decide–act cycle.
5. What does PEAS stand for?
6. Compare BFS and DFS.
7. What is a heuristic? Give an example.
8. Explain knowledge representation and give two forms.
9. Differentiate deduction, induction and abduction.
10. Compare supervised, unsupervised and reinforcement learning.
11. What is overfitting, and how can it be reduced?
12. Explain the roles of training, validation and test datasets.
13. What is a neural network? Name two architecture families.
14. List four NLP tasks.
15. What is computer vision? Give three applications.
16. Define generative AI and explain one limitation of LLMs.
17. What is prompt engineering? List four elements of a useful prompt.
18. Why can accuracy be misleading for an imbalanced dataset?
19. List five principles of responsible AI.
20. Describe a basic workflow for developing an AI system.

14.2 Multiple-choice revision

1. Which is a supervised learning task?
A. Grouping unlabeled songs
B. Predicting house prices from examples with known prices
C. Exploring a maze with no data
D. Manually sorting files

2. Which search strategy expands the lowest-cost path first?
A. DFS
B. UCS
C. Greedy search in all cases
D. Random search

3. What is a feature?
A. An input variable
B. A final exam
C. A type of robot arm
D. A test score only

4. Which metric measures the fraction of predicted positives that are correct?
A. Recall
B. Precision
C. MAE
D. Training loss

5. Which is a common risk of generative AI?
A. It can never make mistakes
B. It always cites real sources
C. It may generate plausible but false information
D. It requires no evaluation

6. Which field focuses on interpreting images and videos?
A. Computer vision
B. Database indexing only
C. Word processing
D. Network cabling

7. In PEAS, the letter S means:
A. Search
B. State
C. Sensors
D. Speed

8. Which practice helps reduce privacy risk?
A. Collect every possible data point
B. Share passwords
C. Minimize data and control access
D. Ignore retention

14.3 Answer guide

1. AI is the field of creating systems that perform tasks associated with learning, reasoning, perception, language or planning.

2. Examples include image recognition, translation, recommendation, prediction and route planning.

3. Traditional programs follow explicitly written rules; ML models learn patterns from examples.

4. An agent observes its environment, chooses an action and acts toward an objective.

5. Performance measure, Environment, Actuators, Sensors.

6. BFS explores shallow levels first; DFS follows a branch deeply before backtracking.

7. A heuristic estimates distance or cost to a goal, such as straight-line distance between cities.

8. KR stores facts and relationships in usable forms such as logic, rules or knowledge graphs.

9. Deduction applies a rule; induction generalizes from examples; abduction proposes a plausible explanation.

10. Supervised learning uses labeled examples; unsupervised learning finds structure without supplied labels; reinforcement learning learns through reward feedback.

11. Overfitting is poor generalization; use more representative data, suitable regularization, simpler models and validation.

12. Training fits parameters; validation tunes choices; test estimates final performance on held-out data.

13. A neural network has connected units and adjustable weights; examples include CNNs and transformers.

14. Examples: translation, sentiment analysis, text classification and summarization.

15. Computer vision extracts information from images/video; applications include detection, OCR and medical-image support.

16. Generative AI creates content; an LLM may hallucinate or reflect bias.

17. Prompt engineering refines instructions; include task, context, audience, constraints and output format.

18. A high score may hide poor performance on a rare but important class.

19. Examples: fairness, transparency, accountability, privacy, safety and human oversight.

20. Define the problem, prepare data, choose/train a model, evaluate, test risks, deploy and monitor.

MCQ answers: 1-B, 2-B, 3-A, 4-B, 5-C, 6-A, 7-C, 8-C.

15. Mini Glossary and Study Plan

15.1 A simple seven-day study plan

Day	Focus	Suggested activity
1	AI basics and history	Learn definitions; write three examples of AI and three examples of ordinary automation.
2	Agents and search	Draw a PEAS table; compare BFS, DFS and UCS.
3	Knowledge and reasoning	Create five IF–THEN rules and label examples of deduction, induction and abduction.
4	Machine learning	Make a table of supervised, unsupervised and reinforcement learning.
5	Neural networks and NLP	Explain a neural network in your own words; list common NLP tasks.
6	Vision, generative AI and workflow	Identify applications and write a responsible prompt.
7	Ethics and revision	Answer practice questions without notes; revisit weak topics.

15.2 How to prepare for an exam

- Memorize accurate definitions, but practise explaining them in your own words.
- For comparisons, write the meaning, one difference and one example for each term.
- For process questions, use ordered steps and a small diagram where useful.

- For application questions, explain the use case, benefit, limitation and mitigation.
- Practise with your course syllabus and lecturer's notes because terminology and depth may differ.

15.3 Final reminder

AI is a broad and fast-changing field. These notes provide a beginner-friendly foundation, but they are not tied to one university syllabus. Always verify course-specific topics, required algorithms and examination expectations with your teacher or official course materials.